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Unlocking The Potential: A Comprehensive Analysis of Tight Carbonate Stimulation Techniques

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Summary

Effective stimulation is a cornerstone in the development of tight carbonate reservoirs, where low permeability and complex heterogeneity severely limit natural hydrocarbon flow. This paper presents a comparative evaluation of five advanced stimulation techniques—Hydraulic Fracturing, Emulsified Acid, Single-Phase Retarded Acid (SPRA), Limited Entry Liner (LEL), and Fishbone Stimulation—implemented across one reservoir having very low permeability in a Middle East field. Each method demonstrated substantial performance enhancements, with injectivity and production gains ranging from 2× to 6×. Hydraulic fracturing proved essential in ultra-tight zones, delivering the highest initial production rates. Fishbone stimulation successfully connected stratified layers, sustaining long-term output. Emulsified acid and SPRA treatments tripled injectivity in water injectors, supporting reservoir pressure and enabling higher oil offtake. LEL stimulation, deployed as a lower completion strategy, ensured uniform acid distribution along long horizontal laterals, overcoming reservoir heterogeneity and enhancing near-wellbore permeability. It delivered significant productivity improvements across various well types, not limited to ESP-equipped producers. Economic analyses revealed rapid payback periods— often within months—and strong net present value (NPV) across all techniques. These results affirm that stimulation is not merely a well enhancement tool but a strategic enabler of full-field development in tight reservoirs. A tailored, reservoir-specific stimulation strategy—integrating mechanical, chemical, and completion-based methods—is essential to unlock deliverability, sustain production, and maximize recovery.

INTRODUCTION

Tight carbonate reservoirs pose significant production challenges due to their extremely low permeability and complex geological heterogeneity. These formations often contain pore systems with micro- to nano-scale throats and are shaped by diagenetic processes such as cementation, compaction, and dissolution, which create isolated zones of porosity and permeability. As a result, conventional production methods are typically uneconomic, and advanced stimulation techniques are required to establish viable flow paths and connect productive zones [1]. The engineering focus shifts from enhancing bulk permeability to strategically accessing and linking these isolated storage domains while avoiding premature breakthrough in high-permeability thief zones.

To address these challenges, stimulation philosophies have evolved from simple near-wellbore treatments to sophisticated technologies that offer precise control over fluid placement and reservoir contact. This paper analyzes five such techniques—Hydraulic Fracturing, Emulsified Acid, SPRA, LEL, and Fishbone Stimulation—each suited to specific reservoir conditions. Their effectiveness depends on a thorough understanding of petrophysical and geomechanical properties, such as pore architecture, natural fracture networks, and stress regimes. In particular, natural fractures and vertical stratification play a critical role in fluid flow, necessitating stimulation strategies that ensure connectivity across layers and effective fluid distribution. The study aims to provide a structured framework for selecting optimal stimulation methods based on reservoir characteristics and field performance.

MATERALS & METHODS

Reservoir Overview

Reservoir R is a tight, lower cretaceous carbonate with modest porosity but very low permeability. The gross thickness is around 80-90 ft, divided into thin sub-zones (P1 through P4 separated by dense stylolitic barriers). Matrix porosity averages ~20% but ranges from as low as 5% in tight streaks up to ~30% in the best facies. Crucially, the matrix permeability is on the order of 0.01 to a few millidarcies - essentially two to three orders of magnitude below what is needed for natural flow. Such pore structure is dominated by micro- to nano-scale throats, a result of depositional mudstone fabrics further compacted and cemented over geologic time. The diagenetic overprinting (e.g. extensive calcite cementation) has created a highly heterogeneous rock fabric where isolated high-porosity vuggy intervals are embedded in a non-permeable matrix. Connectivity is poor, and much of the oil is effectively "trapped" behind capillary forces in micropores. Indeed, reservoir studies indicate significant capillary-bound water and oil: many zones have high irreducible water saturations and low resistivities, consistent with micro-porosity and fine pore throats that impede fluid flow.

Natural fractures do exist - the reservoir is a gently dipping faulted anticline, and core/imagery show some sparse fractures - but their contribution to flow is limited. Fracture porosity is negligible (<0.1%) and the fracture network is not well-connected across the reservoir. Instead, the reservoir behaves like a dual-porosity system where the matrix holds most of the oil and any production must occur via fracture or artificial channels. Additionally, vertical permeability is very low. The reservoir is partitioned into roughly seven depositional layers (including tight separators) with $K_v/K_h \ll 0.5$. Without intervention, a horizontal well in one layer may not drain adjacent layers because fluids cannot move vertically to the wellbore. All these factors - ultra-low matrix k, high capillary entry pressure, and stratified flow units - define the fundamental production challenge: Reservoir R has plenty of oil in place, but an unassisted well cannot effectively deliver it. This harsh reality dictates the need for aggressive stimulation to create or enhance flow paths (fractures, wormholes, etc.) that connect the wellbore to the far-reaching oil-rich pockets in the rock. The stimulation strategy for this reservoir development was therefore predicated on these facts, targeting ways to bypass near-wellbore damage, connect isolated pore networks, and enhance permeability on a reservoir scale.

Stimulation Strategy and Execution

Designing a fit-for-purpose stimulation program for Reservoir R required close collaboration across subsurface disciplines. Geoscientists, reservoir engineers, completion engineers, and production technologists worked as an integrated team to customize the stimulation approach for each well and region of the field. The workflow began with detailed reservoir modeling: the team used static and dynamic models to identify high-priority zones (e.g. thicker, oil-saturated layers like P3) and diagnose the main flow barriers (e.g. specific tight streaks or depletion of certain regions). These insights fed into a decision matrix for stimulation. For example, in areas where modeling showed continuous low-perm rock over long distances, hydraulic fracturing was flagged as the preferred option to create extensive new flow channels. In contrast, for wells intersecting multiple stacked flow units, the model indicated vertical connectivity was the issue, so a fishbone completion (to physically connect layers) was recommended. Conversely, if the issue was primarily near-wellbore skin or damage in an otherwise moderate-perm interval, a chemical stimulation (acidizing) approach was selected.

The stimulation strategy employs a portfolio of advanced techniques each chosen to address specific reservoir challenges. This section summarizes five stimulation techniques used to enhance productivity in Reservoir R as part of its development strategy. Each method is grouped by its core mechanism and described in terms of principle, application, and operational highlights.

Hydraulic Fracturing: Hydraulic fracturing is a high-energy technique used to create new fractures in ultra-tight formations. It involves injecting fluid at high pressure to exceed the formation's fracture gradient, resulting in vertical fractures that improve reservoir connectivity [2].

- Propped Fracturing uses sand or ceramic proppant to keep fractures open.
- Acid Fracturing uses HCl to etch fracture faces, creating conductive channels without proppant.

Operations are staged using the plug-and-perf method, requiring bridge plugs, perforating guns, high-pressure pumps, and coiled tubing for post-job cleanout.

Retarded Acid Systems: These chemical methods improve matrix permeability by slowing acid reaction rates, allowing deeper penetration and better wormholing compared to conventional HCl acid systems as shown in Fig 1.

- **Emulsified Acid:** A water-in-oil emulsion where acid droplets diffuse through an oil barrier. It creates long wormholes and diverts acid into tighter zones [3]. Requires diesel, surfactants, and high-shear mixing.

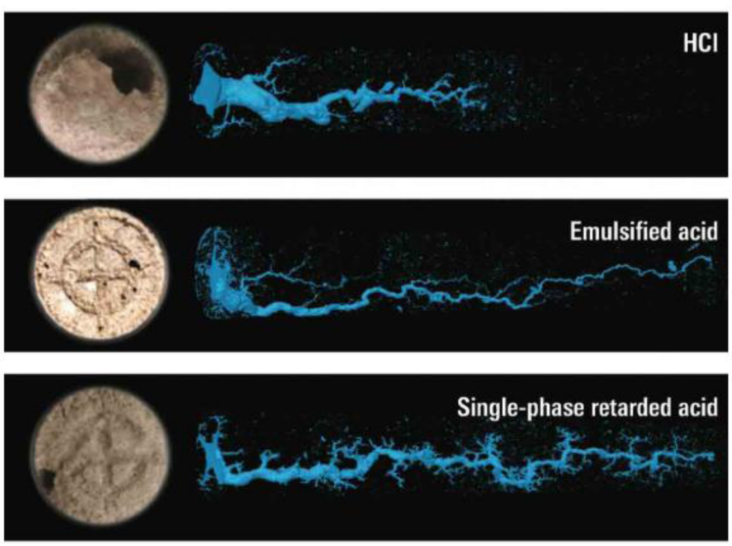

Figure 1—Wormhole propagation in HCl, emulsified acid and Single Phase Retarded Acid (SPRA) in an Indiana Limestone core (Courtesy: SLB)

SPRA (Single-Phase Retarded Acid): Uses chemical retarders in a single-phase acid. It simplifies logistics, reduces friction pressure, and eliminates diesel handling. Ideal for long laterals and high-temperature wells [4].

Mechanical Diversion & Precision Placement

- **Limited Entry Liner (LEL):** Also known as Controlled Acid Jetting (CAJ), a perforated liner is used to control acid distribution along horizontal wells. Engineered hole spacing ensures uniform treatment from heel to toe [6]. Effective in heterogeneous reservoirs and long laterals as shown in Fig.2.
- **Fishbone Stimulation:** Deploys small-diameter laterals ("needles") from the main borehole to connect isolated reservoir layers [5]. Variants include acid jetting and mechanical drilling. Activated by surface pumping, followed by cleanout to ensure full bore access as presented in Fig.3.

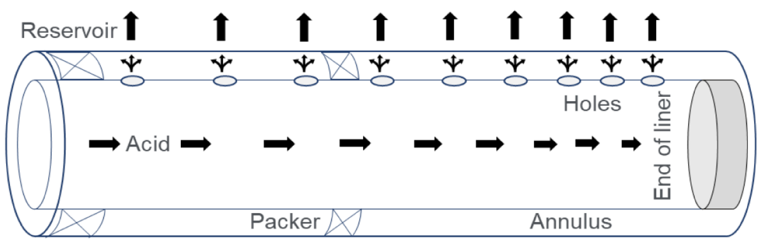

Figure 2—Limited entry liner completion and stimulation concept

Figure 3—Fishbone needles extending into a formation

Table 1 provides a concise, side-by-side comparison of the operational aspects of each technology.

Table 1—Comparison of different stimulation techniques operation

Feature	Hydraulic Fracturing	Emulsified Acid	Single-Phase Retarded Acid (SPRA)	Limited Entry Liner (LEL)	Fishbone Stimulation
Stimulation Principle	Creation of a new, high-conductivity tensile fracture.	Chemical retardation of acid reaction via an oil-external diffusion barrier.	Chemical retardation of acid reaction via a surface-adsorbed chemical film.	Mechanical diversion of fluid via pressure drop across engineered orifices.	Creation of multiple, small-diameter laterals to access the reservoir.
Primary Surface Equipment	High-pressure pumps, blenders, proppant handling, frac tanks.	Acid pumps, blend tanks, high- shear mixers.	Acid pumps, chemical additive skids.	High-rate acid pumps.	Standard rig pumps.
Key Downhole Equipment	Bridge plugs, perforating guns.	None (can use CT).	None (can use CT).	Pre-drilled liner, swellable packers.	Liner with integrated Fishbone subs, fishbasket for cleanup.
Typical Fluid System	Viscous gelled fluid, proppant (sand/ ceramic), flush fluid.	70:30 HCl: Diesel emulsion with surfactants, corrosion inhibitors.	Aqueous HCl with a chemical retarder, corrosion inhibitors.	15% HCl or other stimulation fluid.	15% HCl (jetting)

Feature	Hydraulic Fracturing	Emulsified Acid	Single-Phase Retarded Acid (SPRA)	Limited Entry Liner (LEL)	Fishbone Stimulation
Core Operational Steps	Multi-trip "plug-and- perf" staging, followed by plug mill-out.	Batch mix emulsion, pump treatment, flow back to break emulsion.	Mix on-the-fly or batch mix, pump single-phase fluid.	Run liner, bullhead acid at high rate in a single stage.	Run liner, pump activation fluid in a single stage, run fishbasket.
Key Risks/ Limitations	Screen-out, high operational complexity, large surface footprint, water usage.	High friction pressure, emulsion instability at high temp, requires diesel.	Temperature limits of retarder chemicals, potential for polymer degradation.	Requires high pump rate, erosion of holes, potential for plugging.	Specialized hardware, cost, risk of incomplete needle deployment.

In summary, Reservoir R's stimulation program was not about one technique in isolation but rather using the right tool for the right problem. Table 2 (a conceptual decision matrix) was developed to guide these choices. For example, if a well's target interval had >1 mD permeability and significant skin, LEL acidizing would be primary (SPRA or emulsified acid as the fluid); if the interval had <0.5 mD and homogeneous, fracturing was favored; if the well spanned multiple tight layers, fishbone was chosen in lieu of or in addition to fracturing. This strategic deployment of technologies was backed by meticulous operational execution as described - each method required specific equipment and procedures, all of which were planned in advance. The result was a suite of stimulated wells, each optimally treated to overcome the dominant local flow barrier, whether it was microscopic (pore-scale) or macroscopic (layer- scale).

Table 2—Comparative matrix of different stimulation techniques

Criterion	Hydraulic Fracturing	Emulsified Acid	Single-Phase Retarded Acid (SPRA)	Limited Entry Liner (LEL)	Fishbone Stimulation
Reservoir Applicability					
Optimal Permeability	Very Low (<1 mD)	Low to Medium	Low to Medium	Medium	Low to High
Max Temperature	High	Medium to High	High	Very High	Very High
Suitability for Fractured Reservoirs	Medium (Risk of complexity)	High	High	High	Excellent
Suitability for Layered Reservoirs	Low (Planar fracture)	Medium (Viscous diversion)	Low	Excellent (Mechanical diversion)	Excellent (Physical connection)
Stimulation Efficacy					
Penetration Mechanism	Planar Fracture	Wormhole Network	Wormhole Network	Jetted Wormholes	Discrete Laterals
Typical Penetration Depth	Very Deep (100s ft)	Deep (10s ft)	Deep (10s ft)	Deep (10s ft)	Medium (30-40 ft)
Reservoir Contact Type	Large Planar Area	Volumetric Network	Volumetric Network	Uniform Linear Coverage	Multi-Point Access
Precision/Control	Low	Medium	Medium	High	Excellent
Operational Profile					
Execution Complexity	Very High	High	Low	Low	Medium
Surface Footprint	Very Large	Medium	Small	Small	Small
Key Risks	Screen-out, Frac-hits	Emulsion Stability, High Friction	Chemical Stability	Hole Plugging, Erosion	Hardware Failure
HSE Profile	High (Water, CO_2)	Medium (Diesel)	Low	Low	Very Low
Economic Profile					

Criterion	Hydraulic Fracturing	Emulsified Acid	Single-Phase Retarded Acid (SPRA)	Limited Entry Liner (LEL)	Fishbone Stimulation
Relative CAPEX	Very High	Medium	Low	Medium	High
Relative OPEX	High	Medium	Low	Low	Low
Typical Production Profile	Very High IP, Steep Decline	Good IP, Sustained Decline	Good IP, Sustained Decline	Good IP, Sustained Decline	High IP, Shallow Decline
Overall Economic Potential	High (in tightest rock)	Medium	High	High	Very High (in right context)

Field Trial Results

In this section, we present the outcomes of the different stimulation techniques applied in the Reservoir R and discuss their impacts on well performance. All results are benchmarked against pre-stimulation performance and are analyzed in the context of improving injectivity, productivity, and overall field recovery. Key findings and interpretations are highlighted, and figures and tables are provided to illustrate the performance improvements. The data demonstrate significant enhancement in both injection and production rates, validating the effectiveness of these stimulation methods in the reservoir.

Multi-Stage Hydraulic Fracturing Pilot - Tight Reservoir Stimulation

The most dramatic production gains were achieved with a multi-stage hydraulic fracturing pilot on Well Prod-1, a tight carbonate oil producer. Prior to stimulation, Prod-1 exhibited low natural flow rates (100-300 BOPD) and modest gas-lift assisted rates (~500 BOPD) due to very low permeability (K < 1 mD) along a long horizontal (4,800 ft) in the formation. The well's productivity index (PI) was only ~0.5-0.7 bbl/d/psi, indicating poor efficiency in draining the reservoir. A three-step pilot was executed to fracture stimulate this well:

Fracturing Design and Execution: The fracturing treatment targeted multiple clusters per stage, covering a gross interval of ~1,800 ft with propped fractures (approximately 150 ft half-length and ~60-70 ft height each as per design). A viscous crosslinked gel carrying proppant was used, and acid was also injected as a pad in each stage ("acid touch" stage) to help break near-wellbore tortuosity. A ball- drop system and frac plugs were utilized to isolate stages sequentially. Notably, a mini-frac (diagnostic fracture injection test) was performed first; its data allowed customizing the job to avoid fracturing out of zone, critical in this layered carbonate. Fig. 4 illustrates a representative pressure-time plot for one of the frac stages, showing controlled execution and pad, slurry, and flush sequences. All 12 stages were pumped to completion without screen-out, and the pressures observed aligned with predictions, indicating that the target zones were effectively broken down and propped.

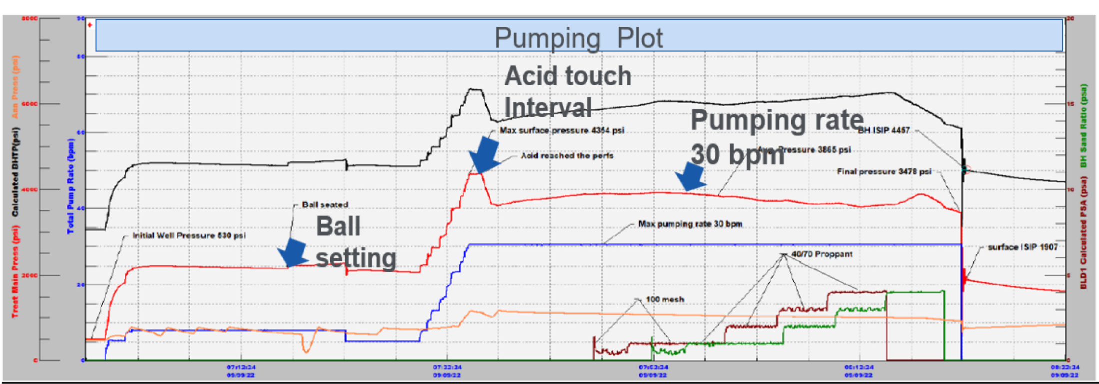

Figure 4—Pressure-time plot during frac operation

Post-Frac Well Performance: The impact on well performance was immediate and substantial. Upon opening the well, Prod-1 flowed naturally at ~1,100 BOPD on a 20/64″ choke - more than 3× its pre-fracturing rate. As the well was gradually beaned up, it reached a peak of ~2,800 BOPD on a 52/64″ choke with 10% water cut. This is nearly a 5–6× increase over the maximum rate with gas lift before the treatment (which was ~500 BOPD). Fig 5 shows the production performance during the initial flowback and testing period.

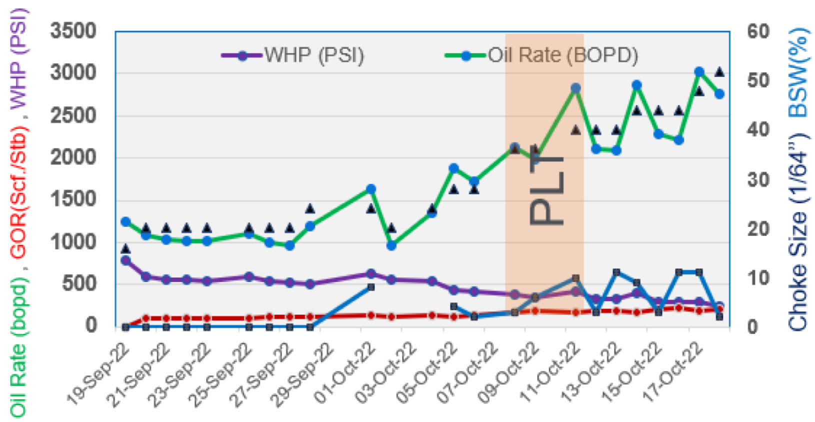

Figure 5—Well performance post Hydraulic Fracturing

The hydraulic fracturing pilot transformed Prod-1 from a marginal producer to one of the best wells in the field. By creating a connected fracture network through the low-permeability rock, the effective well radius and drainage area were massively increased. The PI jumped roughly four-fold, as seen by the post-frac PI of ~2.0 bbl/d/psi compared to ~0.5 before. The positive outcomes also validated the frac design methodology - executing a diagnostic minifrac, adjusting stage volumes, and monitoring pressures closely ensured fracture containment within the desired interval (no signs of out-of-zone growth were observed in offset well pressures or PLT data). Economically, the fracturing approach proved its value. Even though the pilot was capital-intensive, the production gains led to a rapid payback. The comparison with the nearby un-fractured well (40 vs. 275 MBbl in the same period) highlights a 7x increase in recovery over that time frame as it is illustrated in Figure 6.

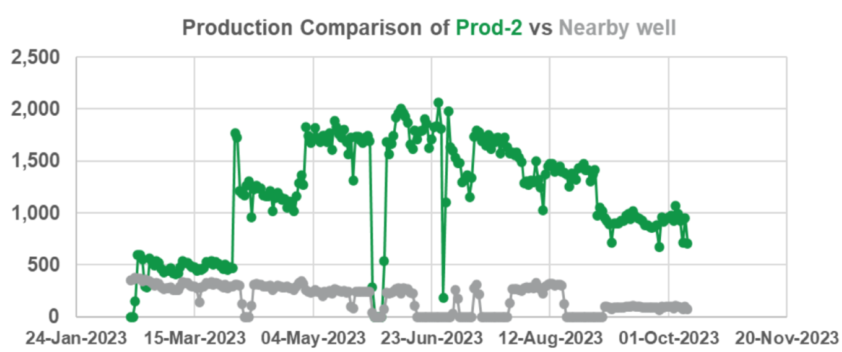

Figure 6—Production Comparison of Prod-1 vs Nearby well post Hydraulic Fracturing

Fishbone Stimulation - Extended Reservoir Contact

Another pilot involved fishbone stimulation technology on horizontal well Prod-2 to improve production from tight layers by increasing reservoir contact and vertical connectivity. Fishbone completion entails running an uncemented liner with multiple fishbone subs (laterals) and then jetting acid to enable the subs' needles to penetrate the formation. In Prod-2, 40 fishbone subs (each with 4 needles of ~40 ft length) were installed along the 4,500-ft horizontal section. After pumping ~3,000 bbl of acid and performing a series of cleaning runs, about 160 needles were successfully deployed into the formation as it shows in Fig 7.

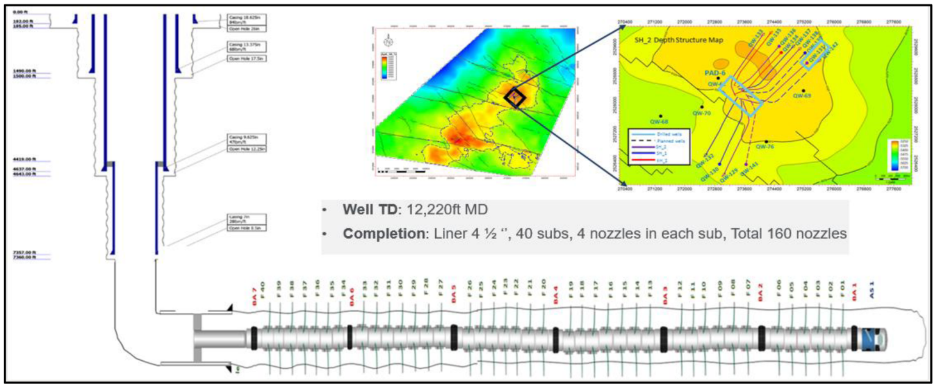

Figure 7—Fishbone penetration Length Inside formation

The fishbone operation was very successful: 70% of the needles achieved at least 50% of their designed penetration length into the reservoir. This corresponds to roughly 3200 ft of new reservoir exposure (in addition to the original borehole length) created by the laterals, dramatically increasing the well's contact area. Table 3 summarizes the distribution of needle penetration lengths achieved in Prod-2 after stimulation. Over 30% of the needles extended beyond 30 ft into the formation (rated "Excellent"), and another ~39% reached between 15-30 ft ("Good"). Only a small fraction (7%) was under 5 ft penetration ("Poor"). Moreover, 100% of the lateral sections were confirmed accessible after the cleaning operations, indicating successful removal of any temporary plugging or debris inside the liner. As summarized in Table 3, over 70% of needles exceeded 15 ft penetration into formation (excellent/good), greatly extending reservoir contact by ~3,200 ft beyond the original wellbore length.

Table 3—Distribution of fishbone lateral (needle) penetration lengths achieved in well Prod-2.

Penetration Length	Needles (count)	Fraction of Total	Assessment
> 30 ft	50	31%	**Excellent**
15–30 ft	63	39%	**Good**
5–15 ft	36	23%	**Average**
< 5 ft	11	7%	**Poor**
Total	**160**	**100%**	

These results translated into a substantial productivity improvement for Prod-2. After the fishbone stimulation, the well's inflow performance was significantly better than nearby conventional wells. Flow rates for Prod-2 pre- and post-fishbone are highlighted in Fig.8, showing that Prod-2's contribution increased markedly. For instance, drawdown requirements dropped due to the larger drainage area (implying a higher Productivity Index, PI) and the well was able to produce at higher rates at lower flowing pressure. In addition, fishbone technology effectively connected discrete sub-layers of the reservoir. By piercing through low-permeability barriers between layers, the laterals established vertical communication that a normal horizontal well could not achieve. This technology is especially suitable for heterogeneous carbonate reservoirs where permeability varies between layers; the fishbone needles bypass nearwellbore damage and connect natural fracture networks that were not in communication with the main bore.

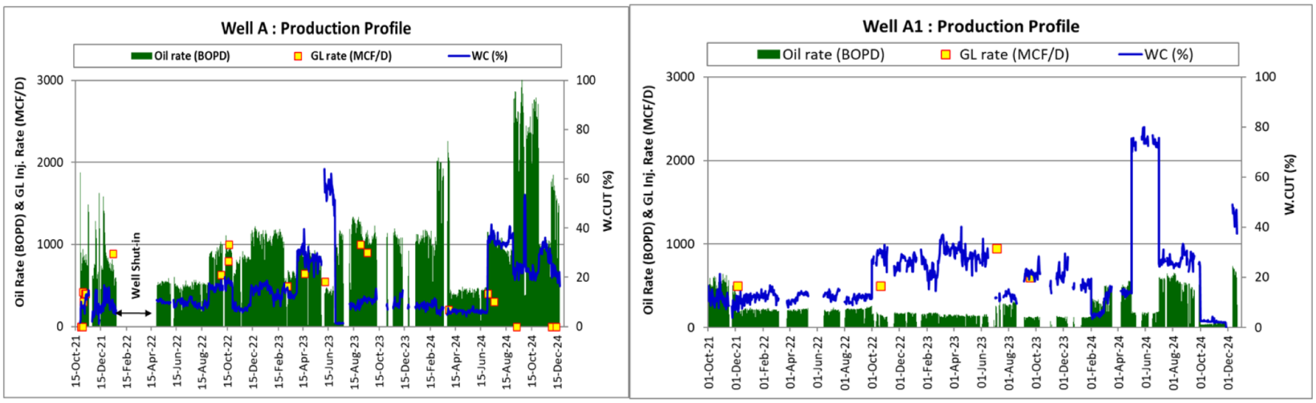

Figure 8—Fishbone Well Prod-2 Vs Conventional Well sustainable performance

Limited-Entry Liner Acid Stimulation

Well Prod-3 was completed with an LEL lower completion for matrix acidizing which was producing around 350 BOPD prior to the stimulation. The LEL completion (a liner with multiple small entry points spaced out) was designed to ensure distributed acid intake along the wellbore. A total of 2,425 bbls of 15% HCl acid (equivalent to 0.7 bbl/ft of lateral length) was bullheaded into the formation. The production rate of Prod-3 approximately doubled from 350 BOPD to ~800 BOPD after the acid stimulation as it shows in Fig.9, with a four-fold improvement in the productivity index (PI) reported. The significant PI gain suggests that skin damage around the well was effectively removed and that the LEL completion distributed acid into multiple zones, not just a single high-perm streak. The wellhead pressure and pump parameters after the job indicated a much lower drawdown was needed to achieve the higher rate, confirming improved near-wellbore permeability.

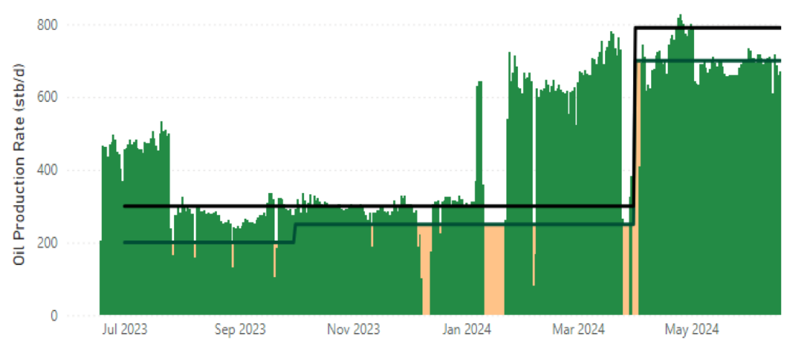

Figure 9—Well performance pre and post LEL Stimulation

Emulsified Acid Trials

Emulsified acid stimulation was piloted on four water injectors to improve injectivity in the Reservoir R which is crucial to provide adequate pressure support for the oil producers. Table 4 summarizes the injection rates before and after treatment for these wells. All four injectors showed **2**.5× to 3.3× increases in water injection rate after treatment. For example, well Inj-1 increased from handling 780 BWPD (barrels of water per day) before stimulation to 2,500 BWPD after emulsified acid treatment, and Inj-2 rose from 1,800 to 6,000 BWPD. These large improvements in injectivity reduced the wellhead pressure required to maintain target rates and helped address voidage replacement ratio (VRR) challenges in the field.

Table 4—Water injectivity before vs. after emulsified acid stimulation in selected injectors

Injector Well	Pre-Stimulation Rate (BWPD)	Post-Stimulation Rate (BWPD)	Improvement
Inj-1	780	2,500	≈3.2 × higher
Inj-2	1,800	6,000	≈3.3 × higher
Inj-3	900	2,500	≈2.8 × higher
Inj-4	600	2,000	≈3.3 × higher

The 20% diesel-based emulsified acid system coupled with high-velocity jetting was successful in bypassing near-wellbore damage and enhancing connectivity to the reservoir's sub-layers**,** as evidenced by the sharp rise in injection capacity. Notably, Inj-1 and Inj-2 were offset injectors paired with the pilot producers (Prod-1 and Prod-2 described earlier), ensuring that the acid stimulation benefits could be observed in corresponding production responses. The trials also utilized extended-reach tools and agitators to ensure acid placement along the lateral extent of the injectors. These results confirm that emulsified acids can effectively improve injectivity in tight carbonates, which is crucial for maintaining reservoir pressure without exceeding fracture pressure limits (-1500 psi wellhead).

Single Phase Retarded Acid (SPRA) Systems

Single Phase Retarded Acid (SPRA) treatments were introduced as a next-generation technique to further improve acid penetration without the need for diesel emulsions. SPRA fluids are single-phase acid systems with chemical retarders that eliminate the use of diesel and provide deeper wormholing than conventional HCl. Selecting the optimal SPRA formulation for a reservoir begins with a thorough understanding of the reservoir's lithology, permeability, porosity, and fluid composition. Engineers initiate the process by conducting laboratory tests—especially Pore Volume to Breakthrough (PVbt)—to evaluate acid propagation and retardation behavior across different rock types. Compatibility tests are then performed to ensure the acid does not cause emulsion or sludge when interacting with reservoir fluids. Based on these results, formulations are shortlisted and tailored to match specific reservoir conditions, including temperature, pressure, and mineralogy.

Water injector Inj-5 had plateaued at ~1000 BWPD injection despite two prior conventional acid stimulations (15% HCl in Jan- 2022 and 28% HCl in Apr-2023). In March 2024, SPRA formulation treatment was performed via coiled tubing. The result was a doubling of injectivity - the well's sustainable injection rate increased from ~1000 to 2,150 BWPD. This two-fold improvement brought Inj-5's injectivity in line with field targets. The success of Inj-5's SPRA (which was bull headed without specialized jetting tools) demonstrates that retarded acids can effectively restore injectivity at lower cost and complexity. Fig. 10 shows the operation parameters during the SPRA job. Importantly, Inj-5's wellhead pressure dropped significantly for the same rate, confirming improved near-wellbore permeability post-SPRA. No adverse effects (such as emulsions or formation plugging) were observed; the injection profile was stable over subsequent months.

Pre Stimulation Injectivity Test. WHCIP = 1200 psi	
Pump Rate	**Pressure**
1.3 bpm	1200 psi
1.6 bpm	1300 psi
1.8 bpm	1500 psi
Post Stimulation Injectivity Test. WHCIP = 1300 psi	
Pump Rate	**Pressure**
2.2 bpm	1300 psi
3 bpm	1500 psi

Figure 10—Inj-5 Pre and post injectivity during the stimulation operation

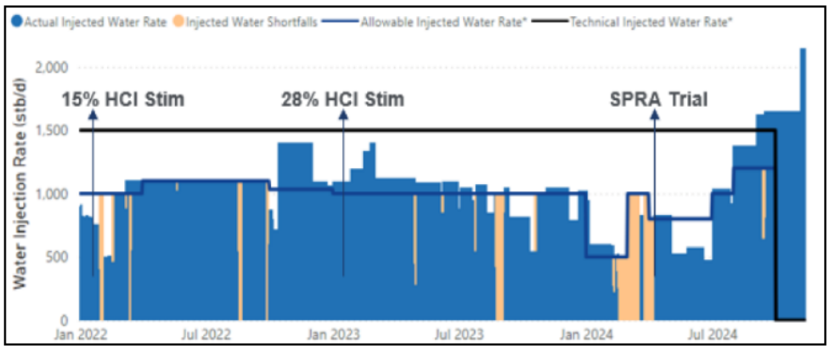

Figure 11—Inj-5 well performance pre and post SPRA Stimulation

The SPRA trials demonstrate the value of diesel-free retarded acid systems as a complementary stimulation approach. Compared to emulsified acid, SPRA fluids offer simpler logistics (single-phase fluid) and lower cost, since they do not require diesel or special mixing equipment. The Inj-5 case confirmed that SPRA can penetrate damage where straight acid could not - likely due to slower spending and deeper wormhole propagation. Achieving a two-fold injectivity increase in a previously "stuck" injector extended the usefulness of that well without expensive recompletion.

Conclusions and Recommendations

The experience in tight carbonate reservoir leads to several key conclusions and lessons learned, which can inform stimulation strategies in similar contexts:

- **Stimulation is Essential for Tight Carbonates:** Conventional development approaches alone (even horizontal wells) were insufficient in Reservoir R due to extreme permeability constraints. Advanced well stimulation proved to be a critical enabler of production - turning non-commercial wells into prolific producers. The need for stimulation in such reservoirs cannot be overstated: it is the difference between recovering a meaningful portion of the oil versus leaving most of it stranded. Any field with matrix permeability in the sub-millidarcy range and poor natural fractures should plan for integrated stimulation as part of the development, not as a contingency. Field's results underscore that with the right stimulation, even a reservoir with <1 mD permeability can achieve substantial flow and recovery.
- **No One-Size-Fits-All - Use a Portfolio of Techniques:** The comparative analysis of technologies shows each has a niche. Hydraulic fracturing offers the largest step-change in productivity for ultra-tight, homogeneous zones, creating long fractures that tap distant oil. However, fracturing provides less control in layered systems and can be complex and costly. Fishbone stimulation excels in stratified or compartmentalized reservoirs, physically connecting layers and enhancing

vertical sweep. It delivered moderate-high production gains with high precision, though it requires specialized hardware and rig time. LEL (Limited Entry) acidizing is a cost-effective method to stimulate long laterals uniformly - ideal for removing skin and mildly improving near-well perms, but less impactful if matrix perm is extremely low (it won't create far-reaching channels like a frac or fishbone). SPRA and Emulsified Acids are invaluable for injectors and matrix treatments, as they greatly extend acid penetration and thus permeability around the wellbore. In reservoir R, all four methods found successful application. The recommendation for similar reservoirs is to maintain a toolkit of stimulation options and apply each where it fits best. A decision matrix (like Table 2 conceptually presented earlier) should be developed per reservoir, considering factors such as layer viscosity, permeability, stress anisotropy, and operational constraints.

- **Early Integration and Planning:** A crucial factor in the operation success was integrating stimulation design from the beginning of the project. By doing so, wells were drilled and completed in a way that facilitated the chosen stimulation - e.g. casing set at appropriate depths, choosing open-hole vs. cemented based on stimulation, installing fiber optics or downhole gauges to monitor treatments, etc. This upfront planning is strongly recommended. It avoids retrofitting problems (which often compromise results) and reduces costs (doing it right the first time).
- **Monitor and Adjust:** Deploy adequate surveillance to understand stimulation effectiveness and reservoir response. Reservoir R team benefited from PLTs, pressure surveys, tracers, and even DTS data to evaluate each stimulation. This feedback loop allowed optimization of subsequent jobs (for example, adjusting acid volumes or stage spacing based on earlier results). It's recommended to perform at least baseline and post-stimulation transient tests (e.g. DFIT, injection fall-off, pressure build-up) to quantify the achieved permeability improvement or fracture characteristics. Continuous reservoir monitoring (pressure, production allocation) is also necessary to ensure that the stimulations are resulting in the intended reservoir-scale effects (e.g. improved voidage replacement, balanced production from layers). If something is off - say a fracture connected an undesired zone - early monitoring will catch it and allow mitigation (like water shut-off or adjusted injection strategy).
- **Economic Justification and Value:** The conducted cases reinforce that advanced stimulation, while expensive per well, is highly cost-effective per barrel. The pilots all yielded positive NPVs and rapid payouts. This is a strong argument to justify stimulation programs to management. One should calculate the unit technical cost (UTC) of adding production via stimulation vs. drilling. In our case, stimulation had UTC on the order of a few dollars per incremental barrel, often an order of magnitude lower than drilling a new well. This kind of analysis is recommended for project sanction: it flips the perspective of stimulation from a cost center to a value creator.

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Abbreviations:

BOPD - Barrels of Oil per Day
BWPD - Barrels of Water per Day
CAJ - Controlled Acid Jetting
CAPEX - Capital Expenditure
CT - Coiled Tubing
DFIT - Diagnostic Fracture Injection Test
DTS - Distributed Temperature Sensing
ESP - Electric Submersible Pump

HCl - Hydrochloric Acid
HSE - Health, Safety, and Environment
Kv/Kh - Vertical to Horizontal Permeability Ratio
LEL - Limited Entry Liner
MBbl - Thousand Barrels (of oil)
NPV - Net Present Value
OPEX - Operating Expenditure
PI - Productivity Index
PLT - Production Logging Tool
PVbt - Pore Volume to Breakthrough
SPRA - Single-Phase Retarded Acid
UTC - Unit Technical Cost
VRR - Voidage Replacement Ratio

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